

thermodynamic efficiency of the human brain. Ever heard of the human brain overheating because the 'exhaust port was clogged with dust'? In trying to bring one up to par with the other, it is useful to try to understand the contrasting features of the two types of 'brains', notwithstanding their similar labels.

- Brains are analogue, robots work on digital electronics
- The brain uses content-addressable memory, while robots use location-addressable memory
- The brain is a huge, parallel processor machine, while robots are individual, independent, and serial unless explicitly constructed in parallel
- The hardware/software distinction is almost impossible to make for the brain, while it is the fundamental constructive principle for a robotic processor
- Biological synapses and nerves are several times more complex than logic gates

Robotics programming languages - early days

Assembly language has been traditionally used to interface certain commands directly at the level of the microprocessor. This decomposes the entire set of available commands into simple building blocks which can be chained together to perform complicated tasks. Such a set of commands is called an instruction set of the assembly language.

LISP, the world's second oldest programming language (after FORTRAN) is an extremely important language from the robotics point of view. The standard Robotics Operating System (ROS) has been partially written in



Source: Wikimedia

A punch card with FORTRAN code entered into it - a relic from the early days of programming.